

An 8×8 Optical Circuit Switch in Thin-Film Lithium Tantalate

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Abstract We demonstrate the first 8×8 electro-optic optical circuit switch (OCS) on thin-film lithium tantalate (TFLT). The photonic integrated circuit comprises twelve 2×2 Mach-Zehnder switches and achieves < 4 ns switching, sub-1-μW static power consumption, 1 dB excess loss per switching stage, and < 0.5 dB output power stability over 1 hour without active feedback. ©2026 The Author(s)

Introduction

The continuous growth in datacenter traffic fueled by cloud computing and artificial intelligence, is driving the need for fast and energy-efficient optical switching [1, 2]. Optical circuit switches (OCS) in integrated photonics offer a promising route to dynamically reconfigure network topologies without the latency and power penalties of optical-electrical-optical conversions. However, the switching speed of existing integrated OCS technologies remains a limitation.

Conventional thermo-optic switches in silicon photonics offer high integration density but are limited to microsecond-to-millisecond response times with significant static power consumption [3]. Micro-electromechanical (MEMS) switches achieve excellent optical performance at large port counts [4] but require hermetic packaging and mechanical release steps that add fabrication complexity. Critically, the slow actuation of both approaches becomes increasingly prohibitive as OCS configurations scale to hundreds of switching elements.

Electro-optic (EO) switching based on the Pockels effect can overcome this speed limitation, enabling nanosecond-scale switching with static power consumption limited to leakage currents only [5]. The thin-film lithium niobate (TFLN) platform has emerged as a leading candidate for such devices, combining a strong electro-optic coefficient with low optical loss and nanophotonic integration [6, 7]. Individual 2×2 MZI-based EO switches on TFLN have been demonstrated with broadband operation and nanosecond response times [8]. However, TFLN devices are known to suffer from DC-drift phenomena: the electro-optic response drifts on long timescales, causing the operating point of Mach-Zehnder-based switches to shift over time [9]. To compensate for this drift, active feedback stabilization loops are required, a critical drawback for many OCS applications.

Thin-film lithium tantalate (TFLT) has recently emerged as an attractive alternative platform, offering an electro-optic coefficient comparable to TFLN while exhibiting significantly slower dielectric

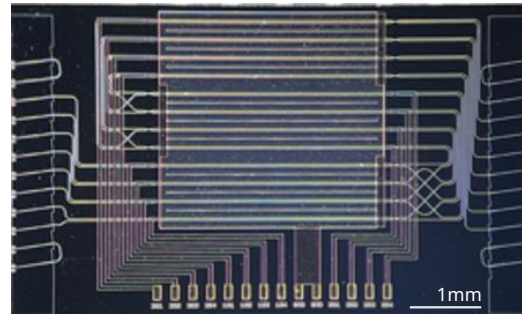


Fig. 1: Microscope image of the 8×8 OCS PIC. The TFLT-based device consists of three cascaded stages of four 3-mm-long MZM switches. The fiber array coupling points are located at the left and right edges of the chip, and the electrical probing pads are located at the bottom edge.

relaxation and, consequently, vastly superior bias stability [10]. Wang et al. demonstrated an individual TFLT 2×2 optical switch with ultralow DC drift [11], but a scalable multi-port OCS on this platform has not yet been reported.

In this paper, we demonstrate the first 8×8 OCS implemented on a thin-film lithium tantalate (TFLT) platform. Comprising cascaded 2×2 Mach-Zehnder modulator (MZM) sub-switches in a 3-stage Banyan network, this device achieves nanosecond switching, sub-1-μW static power consumption, less than 1 dB excess loss per stage, and over 1 hour of output power stability without active feedback.

Technology and Fabrication

The PIC was fabricated on a commercial thin-film lithium tantalate-on-insulator (LTOI) wafer using an advanced pilot line. Electron beam lithography (EBL) was used to define the waveguide structures, which were subsequently transferred into the 400-nm-thick TFLT layer via an optimized reactive-ion etching (RIE) process to ensure low-loss, smooth waveguide sidewalls. Following etching, a top cladding layer of low-loss silicon dioxide (SiO₂) was deposited using plasma-enhanced chemical vapor deposition (PECVD). Finally, a standard lift-off process was employed to pattern the 1300-nm-thick gold contact electrodes.

For optical characterization, two 12-channel single-mode fiber arrays (FAs) were attached to

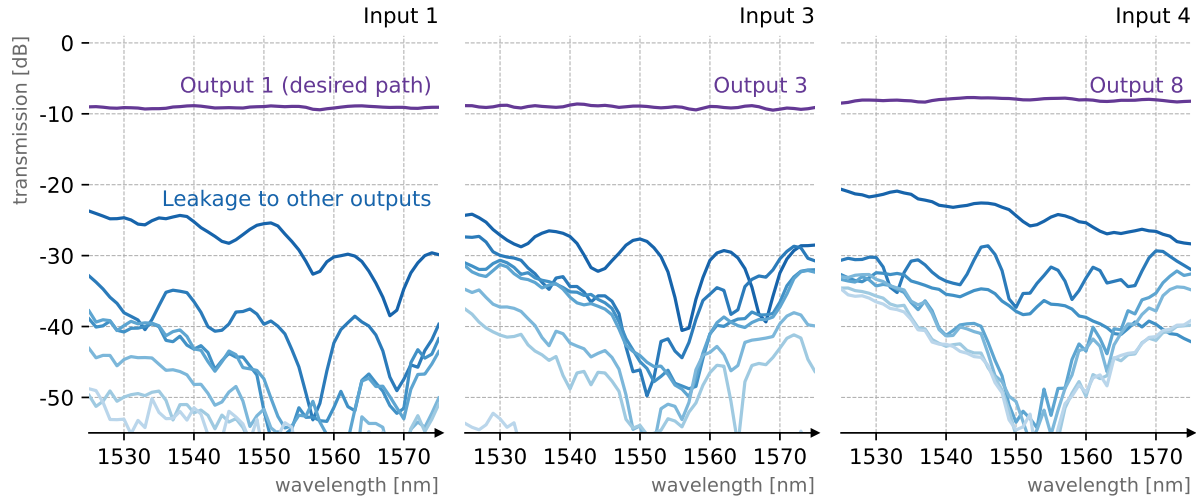


Fig. 2: Measured fiber-to-fiber transmission spectra for three representative switch configurations highlighted in Fig. 3. The desired output path is flat across the C-band with < 1 dB ripple.

achieve robust edge-coupling.

Architecture

The 8×8 OCS is based on fundamental 2×2 Mach-Zehnder Modulator (MZM) switches arranged in a 3-stage Banyan network (Fig. 3), a blocking architecture in which not all input-output permutations can be realized simultaneously. A reconfigurable non-blocking 5-stage Beneš architecture was also fabricated, but not fully characterized at the time of this publication.

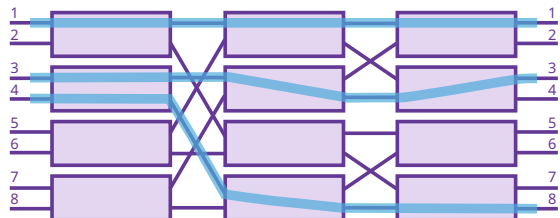


Fig. 3: Schematic of the 8×8 OCS architecture. The device consists of three cascaded stages of four 2×2 MZM switches, configured in a Banyan network topology. The 3 exemplary configurations of Fig. 2 are highlighted in blue.

Individual TFLT Switches

The individual 2×2 switches consist of 3-mm-long TFLT Mach-Zehnder modulators (MZMs) with balanced arms and directional coupler splitters optimized at 1550 nm. The devices are driven in push-pull configuration, applying $-\pi/2$ and $\pi/2$ phase shifts to opposite arms for a full π differential phase shift. We measured a half-wave voltage V_π of 10 V, yielding an effective voltage-length product $V_\pi L$ of 3 V·cm, consistent with reported values for the TFLT platform [10, 11].

Results

The high-speed electro-optic response was characterized by actively toggling a single MZM switch in the chain from the bar to the cross state using a 50 MHz square wave input. Measurements on a

real-time oscilloscope at 50 GS/s revealed switching transients of less than 4 ns (Fig. 4), representing an improvement of several orders of magnitude over thermo-optic and MEMS-based switches [3, 4]. The visible reflection in the transient is a consequence of the unterminated, purely capacitive load presented by the MZM electrodes.

Because the switches rely purely on the Pockels effect without forward current injection or thermal tuning, the static power consumption is negligible. The measured leakage current per switch is in the range of 1.1 to 1.4 nA at the operating voltage of 10 V, resulting in a static power dissipation below 200 nW for all 12 switches in the network.

Figure 2 shows the measured fiber-to-fiber transmission spectra for three representative switch configurations highlighted in Fig. 3: input 1 to output 1 (three bar-state switches), input 3 to output 3 (two cross-state and one bar-state switch, two waveguide crossings), and input 4 to output 8 (one cross-state and two bar-state switches, three waveguide crossings). In all cases, the desired output path exhibits flat transmission across the full C-band (1530–1565 nm) with an intensity ripple below 1 dB.

Systematic loss characterization across 13 different network configurations is summarized in Fig. 5.

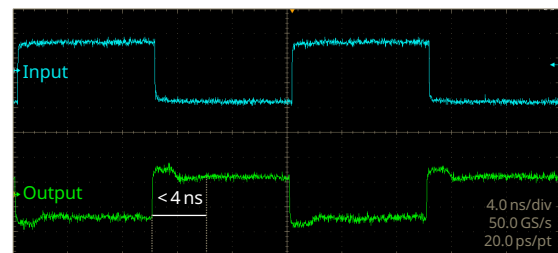


Fig. 4: Switching speed. The transient response of a single MZM switch toggling from bar to cross state is measured using a real-time oscilloscope driven by a 50 MHz square wave input. The observed switching time is less than 4 ns.

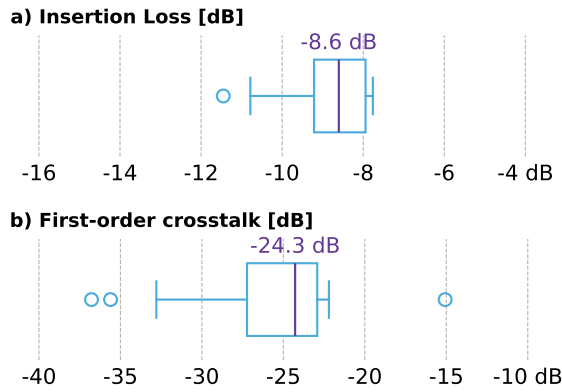


Fig. 5: Statistical analysis (N=13) of the fiber-to-fiber insertion loss (a) and crosstalk (b). The average insertion loss is 8.6 dB and the average crosstalk is -24.3 dB. Only the leakage channel with the highest power, i.e. the first-order crosstalk, is taken into account for the crosstalk analysis.

The average fiber-to-fiber insertion loss is 8.6 dB, of which 6 to 8 dB is attributed to edge coupling (3 to 4 dB per facet). Consequently, a single MZM switching stage contributes only ~ 1 dB of excess loss. Separate test structure measurements indicate that individual waveguide crossings add less than 0.2 dB each.

The average first-order crosstalk at 1550 nm is -24.3 dB (Fig. 5). For this analysis, only the leakage channel with the highest power, i.e. the first-order crosstalk, is taken into account for each input-output pair. Over the full C-band, the worst-case crosstalk degrades to -12 dB at the band edges. We attribute this to the non-ideal splitting ratio of the directional couplers at the input and output of each MZM, which are currently optimized for 1550 nm. Future iterations will replace these with multi-mode interference (MMI) couplers to achieve broadband isolation of 30 dB or more across the full C-band.

To assess the DC-bias stability of the TFLT platform, we conducted repeated continuous 1-hour measurements for various static switch configurations without any feedback stabilization. Figure 6 shows two representative traces at input powers of 10 dBm and 18 dBm. The optical power drift in the intended output channel remained below 0.5 dB over the full hour. The higher-power measurement is particularly noteworthy, as elevated optical power typically exacerbates photorefractive effects in ferroelectric materials [10]. Crosstalk into adjacent channels varied slightly over time but consistently remained below -10 dB across all tested configurations.

Conclusion

We have demonstrated the first 8×8 optical circuit switch fabricated on a thin-film lithium tantalate platform. The device, comprising twelve cascaded 2×2 MZM switches in a 3-stage Banyan

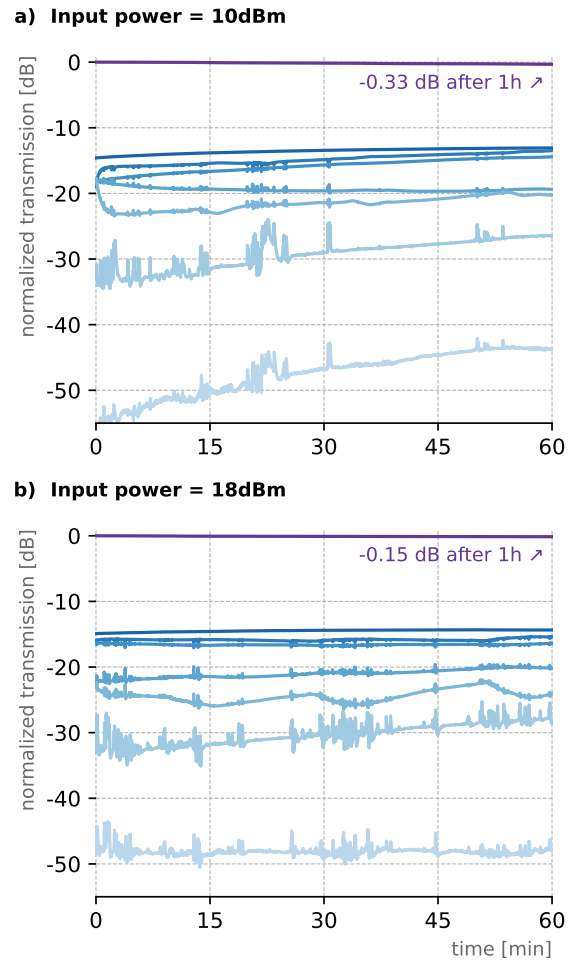


Fig. 6: Long-term output power stability for a static switch configuration measured over 1 hour without active feedback, at input powers of 10 dBm (a) and 18 dBm (b). The intended output channel (dark trace) drifts by less than 0.5 dB. Leakage into unintended channels (blue traces) remains below -10 dB throughout.

architecture, achieves switching transients below 4 ns with static power consumption below 1 μ W. Each switching stage contributes approximately 1 dB of excess loss with flat C-band transmission. Most importantly, the TFLT platform enables passive 1-hour stability with output power drift as low as 0.15 dB. This confirms the significant advantage of TFLT over TFLN for long-term stable OCS operation.

Future work targets broadband MMI-based splitters for improved isolation, full characterization of a non-blocking 5-stage Beneš architecture, and O-band (1310 nm) prototypes for intra-datacenter deployment.

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